

TRAPANI BIRGI, Italy

WMO: 164290

Lat: 37.911N

Lon: 12.488E

Elev: 7

StdP: 101.24

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.1	5.2	0.0	3.8	10.4	1.9	4.3	9.0	14.8	12.9	13.5	13.3	1.6	120	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	34.0	22.4	32.1	23.3	30.8	23.8	27.6	30.0	26.7	29.3	25.9	28.6	6.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.9	22.6	29.6	25.9	21.3	28.8	25.0	20.1	28.0	87.6	30.0	83.6	29.3	79.8	28.8	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.4	9.8	1.8	38.0	1.2	2.0	0.9	39.4	0.2	40.5	-0.5	41.6	-1.4	43.0		
(5)			1.2	29.3	1.2	1.1	0.3	30.1	-0.4	30.8	-1.1	31.4	-1.9	32.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	11.5	11.2	12.9	15.2	18.8	22.6	25.4	26.0	23.6	20.1	16.2	12.9
	DBStd	5.72	2.15	2.27	2.31	2.37	2.60	2.72	2.00	1.95	2.34	2.41	2.51	2.27
	HDD10.0	29	10	12	3	0	0	0	0	0	0	0	0	3
	HDD18.3	952	213	199	169	98	23	1	0	0	0	9	72	167
	CDD10.0	2981	55	46	93	157	273	377	478	497	407	314	187	95
	CDD18.3	861	0	0	1	5	38	128	220	238	157	66	8	0
	CDH23.3	7359	0	0	3	28	239	978	2241	2528	1111	223	7	0
	CDH26.7	2316	0	0	1	5	61	293	766	922	248	21	0	0
Wind	WSAvg	4.0	4.3	4.4	4.4	4.2	3.8	3.4	3.4	3.3	3.6	3.6	4.5	4.5
Precipitation	PrecAvg	477	60	51	44	39	17	4	3	11	45	65	70	68
	PrecMax	875	171	160	134	158	69	23	34	78	153	178	299	192
	PrecMin	200	9	0	0	4	0	0	0	0	0	0	1	2
	PrecStd	149	35	33	32	31	15	5	6	20	37	46	54	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	19.2	22.8	26.3	31.8	35.8	36.2	37.0	33.0	28.8	24.2	20.1
		MCWB	15.2	15.0	15.5	15.4	18.6	21.4	21.9	23.4	22.9	21.6	20.0	17.4
	2%	DB	17.0	17.1	19.9	23.4	28.0	32.2	33.5	33.9	30.8	26.9	22.9	18.5
		MCWB	14.6	14.3	14.9	15.7	17.7	20.7	22.2	23.8	23.7	21.9	19.0	16.1
	5%	DB	16.0	16.1	18.1	21.3	25.9	29.9	31.7	32.0	29.1	25.6	21.5	17.4
		MCWB	13.9	13.7	14.8	15.5	17.7	21.1	23.4	24.7	23.8	21.6	18.2	15.2
	10%	DB	15.1	15.1	17.0	19.8	23.9	27.8	30.1	30.5	27.9	24.4	20.2	16.7
		MCWB	13.1	12.9	14.5	15.4	18.1	21.6	24.2	25.1	23.5	20.8	17.3	14.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	15.9	17.4	19.1	21.9	26.5	28.5	29.2	27.1	24.3	21.6	17.7
		MCDB	17.3	18.0	19.7	22.0	24.8	29.3	30.5	31.6	29.5	26.6	23.5	19.3
	2%	WB	14.9	14.8	16.1	17.6	20.8	24.9	27.4	27.9	25.9	23.1	19.7	16.5
		MCDB	16.5	16.6	18.2	20.6	23.9	27.5	29.7	30.5	28.4	25.5	21.8	18.0
	5%	WB	14.2	14.0	15.3	16.7	20.0	23.9	26.5	27.0	24.9	22.2	18.7	15.6
		MCDB	15.7	15.7	17.4	19.3	23.1	26.8	29.0	29.7	27.8	24.6	20.8	17.1
	10%	WB	13.4	13.2	14.6	15.9	19.1	23.0	25.7	26.1	24.1	21.4	17.8	14.9
		MCDB	14.9	14.8	16.6	18.6	22.3	26.1	28.3	29.0	26.8	23.8	19.9	16.3
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.8	7.5	8.2	8.7	9.2	9.0	8.9	7.9	7.7	6.7	6.2
		MCDBR	7.0	7.6	9.2	11.2	12.2	12.8	12.2	11.3	9.3	8.4	7.4	6.8
		MCWBR	5.6	6.1	6.6	6.5	6.4	7.1	7.8	7.7	6.0	5.6	5.4	5.3
	5% WB	MCDDBR	6.5	7.1	8.4	9.1	10.0	9.8	9.2	9.3	8.3	7.9	7.0	6.1
		MCWBR	5.6	6.0	6.7	6.5	6.9	7.5	7.7	7.8	6.2	6.0	5.6	5.2
Clear-Sky Solar Irradiance	taub	0.349	0.370	0.412	0.451	0.457	0.431	0.438	0.435	0.440	0.416	0.383	0.356	
	taud	2.452	2.373	2.245	2.152	2.179	2.299	2.292	2.305	2.290	2.330	2.370	2.425	
	Ebn at Noon	826	852	845	831	831	852	843	837	808	789	773	786	
	Edh at Noon	84	103	128	149	147	131	131	126	121	106	91	81	
All-Sky Solar Radiation	RadAvg	2.23	3.09	4.43	5.73	6.82	7.64	7.71	6.78	5.08	3.70	2.55	2.00	
	RadStd	0.16	0.28	0.35	0.36	0.36	0.35	0.22	0.16	0.26	0.22	0.23	0.15	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	-0.45	-0.55	N/A	N/A	N/A	N/A

Nomenclature: See separate page